Lichen Huang

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EDUCATION

Cornell University, Ithaca, NY — Aug 2024 - May 2027

- GPA 3.851/4.3 | B.A. in Chemistry, Computer Science, Mathematics, Honor track for Chemistry
- Award: The Darryl H. Wu Award, Department of Chemistry and Chemical Biology, Cornell University | May 2026

PUBLICATIONS

Chengyi Fan, **Lichen Huang**, and Tristan Lambert. Metal-Free Ring Closing Metathesis. ***(Manuscript in Preparation)***

RESEARCH EXPERIENCE

Lambert Group, Cornell University — Jan 2025 - Present

Undergraduate Research Assistant, advised by Prof. Tristan H. Lambert

DFT-Based Machine Learning Model Towards Catalytic Structure Prediction

This work aimed to build an effective machine learning model with limited data to predict effective organocatalyst structures for metathesis, based on previously working structures.

- Developed a molecular structure generator to systematically construct chemically feasible organocatalyst candidates based on established catalyst scaffolds.
- Designed pre-computation algorithms and scripts to automatically globally optimize the geometry of different states and generate atomic coordinates before DFT, saving DFT optimization time and effort.
- Built machine learning models based on random forest models to predict energy properties with generated structures based on designed molecular descriptions by learning from computation results.

Metal-Free Ring-Closing Metathesis (RCM) Enabled by a Novel Organocatalyst

This work developed a metal-free approach to ring-closing metathesis (RCM) using an organocatalyst, providing an alternative to conventional Ru- and Mo-based catalysts.

- Designed and synthesized organocatalysts and contributed to establishing the reaction scope; prepared over 40 substrates and corresponding products and fully characterized the isolated compounds by $^{1}H/^{13}C$ NMR spectroscopy and DART-MS, generating publication-quality scope and characterization data for a forthcoming manuscript.
- Developed and optimized second-generation reaction conditions that improved reaction yields and expanded the substrate scope beyond that accessible under the original conditions.
- Optimized experimental protocols for substrate preparation and developed an efficient microscale purification procedure, allowed for streamlining product isolation, and reduced workload for each substrate by one third.

New Organocatalyst Scaffold Development for Ring-Closing Metathesis

Built on our previous metal-free RCM methodology, this project aims to design and develop a structurally distinct diaryl organocatalyst scaffold to improve and expand catalyst performance.

- Designed and synthesized a series of diaryl organocatalysts, incorporating structural modifications intended to leverage noncovalent interactions and optimize catalytic performance.
- Established proof-of-concept reactivity of the new catalyst scaffold in ring-closing metathesis and optimized

reaction conditions using model substrates.

- Contributed to mechanistic studies by isolating and characterizing key reaction intermediates, providing experimental evidence for two competing mechanistic pathways within the new catalyst family.

INTERNSHIP

Global InfoTech Summer 2025

Software Engineering Intern

- Developed backend features for internal staff portals using Java, Spring Boot, Maven, and REST/Swagger APIs, including new search functionality for querying and filtering client data.
- Designed and implemented DTOs and data-transfer logic to support new backend features and interactions between application components.
- Worked extensively with MySQL, writing complex SQL queries involving multiple-table joins and adding database relationships to support new client-data functionality.
- Followed a commercial software-development workflow involving Git, code review, unit/integration testing, CI/CD pipelines, logging and documentation, gaining experience developing and testing software within a large engineering team.

SKILLS

- **Organic Synthesis**: multi-step small molecule synthesis, synthetic route design and optimization, reaction development and condition optimization, air- and moisture-sensitive techniques, vacuum-line techniques, solvent purification and degassing techniques, solid-state reaction, enzyme catalyzed reactions
- **Purification and Characterization:** Column chromatography, Trituration and recrystallization, HPLC, GC; ^{1}H, ^{13}C, and 2D NMR spectroscopy; IR/FTIR, DART-MS, MALDI-TOF MS, UV–Vis, fluorescence, and gamma spectroscopy; Stern–Volmer analysis and cyclic voltammetry
- Computational Chemistry: DFT calculations using Gaussian, homemade structure generator to generate chemically feasible catalyst structures, homemade script for pre-computation on geometric and global optimization before DFT, machine learning techniques for catalyst design (SVM, neuro networks, random forests, transformers…); homemade pipelines to incorporate different sections
- Software: ChemDraw, MestReNova, Igor Pro, Gaussian, Mathematica
- Programming: Python, Java, C/C++/C#, SQL, Julia, OCaml, Rocq, Lean, Assembly (RISC-V), Harmony, Bash, Mathematica, Igor, HTML, XML, YAML, TypeScript, CSS, GDScript, LaTeX